FIITJEE

ALL INDIA TEST SERIES

FULL TEST – VII

JEE (Advanced)-2021

PAPER –1

TEST DATE-30-05-2021

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

SECTION – A

1. B

Sol. $\frac{q_1}{r}=\frac{Q'}{R}$

$q_1+Q'=Q$

$Q'\left(\frac{r}{R}+1\right)=Q$

$Q'=\frac{QR}{R+r}, q_1=\frac{Qr}{R+r}$

Similarly $Q''=\frac{QR^2}{(R+r)^2}, q_2=\frac{QrR}{(R+r)^2}$

$\frac{q_2^2}{q_1}=\frac{\left[\frac{QrR}{(R+r)^2}\right]}{\left(\frac{Qr}{R+r}\right)}=\frac{QR^2r}{(R+r)}=q_3$

2. B

Sol. $\phi=\frac{q_{net}}{\in_0}$

For a charge placed at corner of cube $\phi = \frac{q_{net}}{8 \in_0}$

Given system

$$\phi = \frac{(1-2+3-4-6+7+5-8)}{2 \in_0} \Rightarrow \phi = \frac{-1}{2 \in_0}$$

3. D

Sol. Consider system of two disks to be short dipole

$$dE = \frac{2kdp}{r^3}$$

$$dp = dp\ell = \sigma dA\ell$$

$$E = \frac{2k}{r^3}\int dp = \frac{2k}{r^3}\ell\sigma A$$

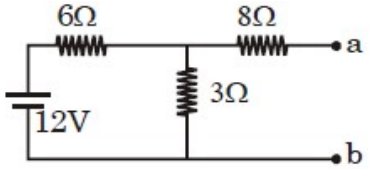
$$E = \frac{2\ell}{4\pi \in_0 r^3}\sigma\pi R^2 = \frac{\sigma\ell R^2}{2 \in_0 r^3}$$

4. B

Sol. $P_{\max} = \frac{V_{ab}^2}{4R_{ab}}; P_{\max} = \frac{16}{4\times 10} = 0.4\text{watt}$

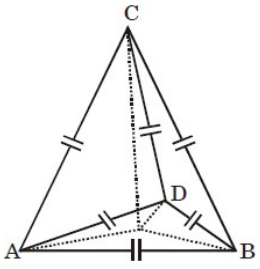
$$V_{ab} = \frac{12}{9}\times 3 = 4\,\text{volt}$$

$$R_{ab} = 8 + \frac{6\times 3}{9} = 8+2 = 10\Omega$$



5. D

Sol. It can be worked out that the circuit diagram is a regular tetrahedron, whose edges contain 6 identical capacitors. From symmetry consideration, irrespective of the pair of points between which the source is connected, there always exists uncharged capacitor in the circuit.



6. C

7. ABD

Sol. Perimeter is decreasing at a rate of 2v

$$\therefore \frac{d}{dt}(2\pi r) = 2v \Rightarrow \frac{dr}{dt} = \frac{v}{\pi}$$

$$\therefore r = \left(r_0 - \frac{v}{\pi} t \right)$$

$$\phi = B \cdot \pi r^2 \Rightarrow \varepsilon = \left| \frac{-d\phi}{dt} \right| = B \cdot 2\pi \cdot r \frac{dr}{dt}$$

$$\therefore \varepsilon = 2B\pi \left(r_0 - \frac{v}{\pi} t \right) \cdot \frac{v}{\pi} = 2Bv \left(r_0 - \frac{v}{\pi} t \right)$$

$$I = \frac{\phi}{R} = \frac{2Bvr}{2\pi r} = \frac{Bv}{\pi\lambda}$$

8. BD

Sol. $(\mathrm{A})\ \mu mg = m\omega^2 r$

$$= \alpha^2 r t^2$$

$$t = \sqrt{\frac{\mu g}{\alpha^2 r}}$$

$(\mathrm{C})\ N = mr\alpha$

$$\mu N = m\omega^2 r$$

$$\mu mr\alpha = m\omega^2 r$$

$$= \alpha^2 t^2$$

$$t = \sqrt{\frac{\mu}{\alpha}}$$

9. AD

Sol. $v_m = v_{mp} + v_p$

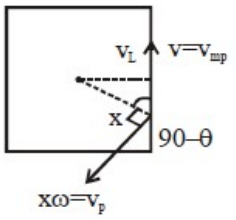
$$= \sqrt{v^2 + x^2\omega^2 + 2v\omega} \cos(90 + \theta)$$

$$= \sqrt{v^2 + x^2\omega^2 - 2vx\omega \sin\theta}$$

$$L_i = 0 = L_f$$

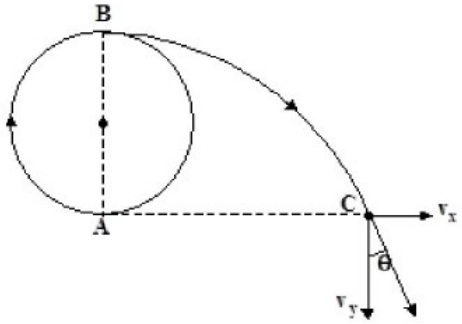
$$= -\frac{m}{12}\left(a^2 + a^2\right)\omega + m\left(v - x\omega\sin\theta\right)\frac{\alpha}{2}$$

$$\Rightarrow \frac{a^2}{6}\omega + \frac{a^2}{4}\omega = \frac{va}{2} \Rightarrow \omega = \frac{6v}{5a}$$



10. AC

Sol. $AC =$ (horizontal velocity at B) (time taken by the body to cover displacement AB) $= \left(\sqrt{gt}\right)\left(\sqrt{\frac{4l}{g}}\right) = 2l$



At point C $v_x = \sqrt{gl}$ $v_y = \sqrt{4gl}$

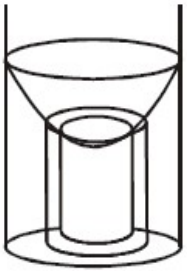
$\tan\theta = \frac{v_x}{v_y} = \frac{1}{2}$

Tangential acceleration $= g\cos\theta - \frac{2}{\sqrt{5}}g$

11. AC

12. AD

Sol. $2h = h + \frac{\omega^2 R^2}{2g} \Rightarrow \sqrt{\frac{2gh}{R^2}} = \omega$



$dK = \frac{1}{2}\times dm\omega^2 x^2 = \frac{1}{2}\rho\times 2\pi x dx\left(h + \omega^2\frac{x^2}{2g}\right)\times\omega^2 x^2$

$K = \pi\rho\omega^2\left[\frac{R^4}{2}h + \frac{\omega^2}{2g}\times\frac{R^6}{6}\right] = \pi\rho\omega^2\left[\frac{R^4}{2}h + \frac{2gh\times R^4}{12g}\right] = \frac{5\pi\rho\omega^2 R^4 h}{6}$

SECTION – C

13. 00003.00

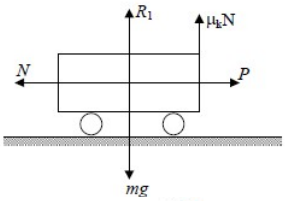
14. 00002.00

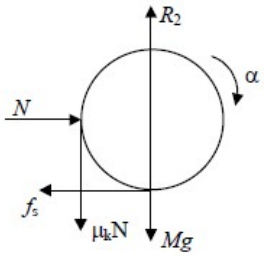
15. 00002.00

Sol. F.B.D. of the block B

$P - N = ma ------(1)$

F.B.D. of the cylinder





$N = f_s = MA_{cm}$

$Nr - \mu_k Nr = \left(I_{cm} + Mr^2\right)\alpha \quad \left[\because A_{cm} = a\right]$

$N = \frac{3Ma}{2\left(1-\mu_k\right)} = \frac{3\times 2\times a}{2\left(1-\frac{1}{2}\right)} = 6a$

Hence, from equation (1), we have

$P = N + ma = \left(6+m\right)a = 16a$

$= 16\times r\times\alpha \quad \left[\because A_{cm} = r\alpha = a\right] \quad = 16\times\frac{1}{10}\times 20 = 32N$

16. 00001.00

Sol. $0.4 \times k_0 = 10$

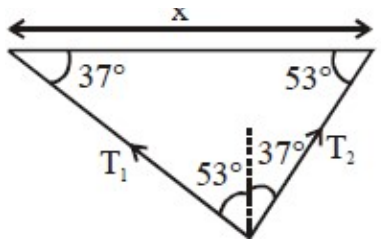
$\Delta\ell = 3k_0 = 30$

$\Delta\ell = 6 \times 0.4 = 0.4\,mm \Rightarrow 1$

17. 00008.00

Sol. $T_2 \sin 37 = T_1 \sin 53$

$3T_2 = 4T_1$



$\ell_2 = x\cos 53° = \frac{3x}{5} \Rightarrow \ell_1 = x\cos 37° = \frac{4x}{5}$

$t_2 = \frac{5}{\sqrt{\frac{T_2}{\mu}}} = 3\sqrt{3} \times 10^{-3} \Rightarrow \frac{3x}{5}\sqrt{\frac{3\mu}{4T_1}} = \sqrt{3} \times 10^{-3}$

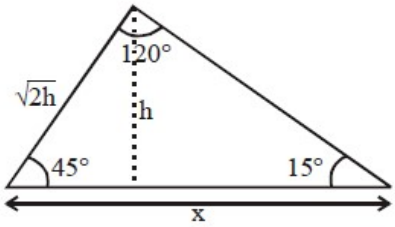
$t_1 = \frac{\frac{4x}{5}}{\sqrt{\frac{T_1}{x}}} = \frac{4}{5} \times \sqrt{\frac{\mu}{T_1}} = \frac{4}{5} \times 80 \times 10^{-3} = 8 \times 10^{-3}$ sec

18. 00005.00

Sol. $\frac{2\sqrt{2} \times \sqrt{2} \times 1.05}{\sqrt{3}-1} = \frac{2x}{\sqrt{3}}$

$\frac{2.1\sqrt{3}}{\sqrt{3}-1} = x$

$x = \frac{2.1}{1-0.577} = \frac{2.10}{0.42}$



Chemistry

PART – II

SECTION – A

19. C

20. C

21. B

22. D

23. C

Sol. Formation of (B) is via intramolecular nucleophilic substitution reaction.

24. A

Sol. The colour of HgI_2, HgO, Pb_3O_4 and $(NH_4)_2Cr_2O$ are red, red, red and orange spark respectively A, B and C all these are insoluble in water. But, all of these becomes soluble if some KI is added to water due to the formation very stable complex HgI_4^2 and PbI_2. But, when the rad solid is heated in a test tube it cause the appearance of Hg metal on the cooler parts of the test tube which suggest that the red solid may be HgI_2 or, it may be HgO. But, on heating it produce violet fumes of I_2 which suggest that the red solid is HgI_2. The reaction are as follows:

$HgI_2 + 2KI \text{®} K_2HgI_4\,(\text{soluble in water})$ $\qquad$ $HgI_2 \xrightarrow{\text{Heat}} Hg + I_2\,(\text{Violet})$

25. ABC

26. ABCD

27. ABCD

28. BCD

Sol. In option A→ along with these Mn also show this property

29. ABCD

30. ACD

Sol. $\dfrac{r_{vap.}}{l_{O_2}} = \sqrt{\dfrac{32}{M_{vap.}}} \Rightarrow \left(\dfrac{4}{3}\right)^2 = \dfrac{32}{M_{vap.}}$

$M_{vap.} = 18$

$\Rightarrow V.D = 9 \Rightarrow Density = \dfrac{18}{22.4}\, gram/ltr$

$= 0.8035\, gram/ltr$

Now, $Z = \dfrac{PV}{nRT} = \dfrac{P}{\dfrac{W}{MV}RT} = \dfrac{PM}{dRT} \Rightarrow Z = \dfrac{18P}{dRT}$

SECTION – C

31. 00001.75
Sol. Total 11 and 2,3,7,8,9,10 and 11 can show

32. 00000.14
Sol actually it is -0.14

33. 00000.45
Sol. $AgBr(s) \rightleftharpoons Ag^+ + Br^-$

x x

$Ag^+ + 2S_2O_3^{2-} \rightleftharpoons Ag(S_2O_3)_2^{3-}$

x- y 0.1 – 2x x

$AgBr(s) + 2S_2O_3^{2-} \rightleftharpoons Ag(S_2O_3)_2^{3-} + Br^-$

$(0.1-2x)$ x x_2

$$K_f \times K_{SP} = 5\times10^{13}\times5\times10^{-13} = \frac{x^2}{(0.1-2x)^2}$$

$$5 = \frac{x}{0.1-2x} \Rightarrow x = 0.045M$$

34. 00025.16
Sol. Given data show ZnS type crystal

$$\therefore PF = \frac{\pi}{3\sqrt{2}}\left[\left(\frac{r^+}{r^-}\right)^3 + 1\right]$$

$$= \frac{3.14}{3\sqrt{2}}\left[(0.225)^3 + 1\right] \text{ since} \frac{r^+}{r^-} = 0.225$$

$$= \frac{3.14\times1.0114}{3\times1.414} = 0.7486$$

% of void=100-74.46=25.15%

35. 00000.11
Sol. $NH_4Cl \rightarrow NH_4^+ + Cl^-, NH_4^+ + H_2O \rightarrow Nh_4OH + H^+$

$\overset{C}{C} - C\alpha$ $C\alpha$ $C\alpha$ $\overset{C\alpha}{C}\alpha - C\alpha h$ $C\alpha h$ $C\alpha h$

$$i = \frac{C - C\alpha + C\alpha - C\alpha h + C\alpha h + C\alpha h}{C}$$

$$= (1 + \alpha + \alpha h)$$

$$\Delta T_f = i \times K_f \times m$$

$$0.637 = \frac{(1+\alpha+\alpha h)\times1.86\times10}{53.5}$$

Since $\alpha = 0.75, h = 0.109$

36. 01659.90

Sol. $\frac{d\ln P}{dT}=\frac{\Delta H}{RT^2}$ ..... (i)

$$\log P=3.54595-\frac{313.7}{T}+1.40655\log T$$

$$\ln P=3.54595-\frac{313.7}{T}\times 2.303+1.40655\ln T$$

$$\frac{d\ln P}{dT}=\frac{313.7\times 2.303}{T^2}+\frac{1.40655}{T}\quad .....(ii)$$

Comparing eqs. (i) and (ii)

$\Delta H=1659.9Cal$

Mathematics

PART – III

SECTION – A

37. C

Sol. $f(x)=\sin x+\int_{-\pi/2}^{\pi/2}\left(\sin x+t\,f(t)\right)dt=\sin x+\pi\sin x+\int_{-\pi/2}^{\pi/2}t\,f(t)\,dt$

$\therefore\ f(x)=(\pi+1)\sin x+A\quad .....(1)$

Now $A=\int_{-\pi/2}^{\pi/2}\left((\pi+1)\sin t+A\right)dt$

$=2(\pi+1)\int_{0}^{\pi/2}\underset{(I)}{t}\ \underset{(II)}{\sin t}\ dt$

$\Rightarrow A=2(\pi+1)$

38. A

Sol. $1=\int_{2\pi/3}^{\pi}\left|2\cos x+1\right|dx=-\int_{2\pi/3}^{\pi}(2\cos x+1)\,dx$

$=2\sin x+x\Big|_{\pi}^{2\pi/3}=\sqrt{3}+\frac{2\pi}{3}-\pi$

$=\sqrt{3}-\frac{\pi}{3}$

39. D

40. D

Sol. $3n-2=7m+2$

$n=\frac{7m+4}{3}$

$n=\frac{7(3\lambda+2)+4}{3}$

$n=7\lambda+6$ and $m=3\lambda+2$

$3\lambda+2\le 2004$ and $7\lambda+6\le 2004$

$\lambda\le 667$ and $\lambda\le 285$

$\Rightarrow 286$

41. C

42. B

Sol. Since $\frac{1}{r^1}+\frac{1}{r^2}+\frac{1}{r^3}=\frac{1}{r}$

So, that r is the smallest term in G.P.

suppose $A\le B\le C$

$\therefore r_1\le r_2\le r_3$

Hence $t>1$, such that $r_1=tr,\ r_2=rt^2,\ r_3=rt^3$

Also, $\Delta=rs=r_1(S-a)=r_2(S-b)=r_3(S-c)$

$\therefore (S-b)=t(S-c),\ (S-a)=t^2(S-c)$

$S=t^3(S-c)$

$\Delta rS=\Delta=\sqrt{S(S-a)(S-b)(S-c)}=t^3(S-c)^2$

Also, $rs=t^3r(S-c)$

$\Rightarrow r=S-c\ so,\ \tan\dfrac{C}{2}=\dfrac{r}{S-c}=1$

$\therefore C=90°$

43. ABCD

Sol. $x^3-x+1=0$

$\Sigma\alpha=0$

$\Sigma\alpha^3=3\alpha\beta\gamma=-3$

$\alpha^2=1=\dfrac{1}{\alpha}$

$\Sigma\alpha^2=3-\Sigma\dfrac{1}{\alpha}\Rightarrow\Sigma\alpha^2=2$

If $S_n=\alpha^n+\beta^n+\gamma^n\Rightarrow S_{n+3}=S_{n+1}-S_n$

$S_4=S_2-S_1=2$

$S_5=S_3-S_2\Rightarrow S_5=-5$

44. AB

Sol. $|A-xI|=0\Rightarrow\begin{vmatrix}2-x & 1 & 3\\ 1 & 1-x & 2\\ 3 & 1 & 1-x\end{vmatrix}=0$

$x^3-4x^2-7x=-3$

$\Rightarrow A^3-4A^2-7A=-3I$

$A\left\{-\dfrac{1}{3}A^2+\dfrac{4}{3}A+\dfrac{7}{3}I\right\}=1$

$A^{-1}=-\dfrac{1}{3}A^2+\dfrac{4}{3}A+\dfrac{7}{3}I$

$\alpha=-\dfrac{1}{3},\beta=\dfrac{4}{3},\gamma=\dfrac{7}{3}$

45. AD

Sol. f(x) is continuous at x = 0 and f′(0) = 0

$\Rightarrow \lim\limits_{h\to0} f(0+h)=\lim\limits_{h\to0} f(0-h)=f(0)=0$

and L.H.D at x = 0 is $\lim\limits_{h\to0}\dfrac{f(0+h)-f(0)}{h}=0$

and R.H.D at x = 0 is 0.
$\Rightarrow$ L.H.D = R.H.D at x = 0 $\quad\Rightarrow f'(0) = 0$

46. ABCD

Sol. Since $\vec{a}, \vec{b}, \vec{c}$ are linearly independent $\quad\Rightarrow \left[\vec{a}\,\vec{b}\,\vec{c}\right] \neq 0$

$\Rightarrow \vec{d}$ can be written as linear combination of $\vec{a}, \vec{b}$ and $\vec{c}$

$$\vec{d} = \frac{\left[\vec{d}\,\vec{b}\,\vec{c}\right]\vec{a} + \left[\vec{d}\,\vec{c}\,\vec{a}\right]\vec{b} + \left[\vec{d}\,\vec{a}\,\vec{b}\right]\vec{c}}{\left[\vec{a}\,\vec{b}\,\vec{c}\right]}$$

Now, $\vec{r} = \left[\vec{d}\,\vec{b}\,\vec{a}\right]\vec{c} - \left[\vec{d}\,\vec{b}\,\vec{c}\right]\vec{a} + \left[\vec{d}\,\vec{c}\,\vec{b}\right]\vec{a} - \left[\vec{d}\,\vec{c}\,\vec{a}\right]\vec{b} + \left[\vec{d}\,\vec{a}\,\vec{c}\right]\vec{b} - \left[\vec{d}\,\vec{a}\,\vec{b}\right]\vec{c}$

$= 2\left\{\left[\vec{d}\,\vec{c}\,\vec{b}\right]\vec{a} + \left[\vec{d}\,\vec{a}\,\vec{c}\right]\vec{b} + \left[\vec{d}\,\vec{b}\,\vec{a}\right]\vec{c}\right\} = -2\left[\vec{a}\,\vec{b}\,\vec{c}\right]\vec{d} \quad\Rightarrow \vec{r} = -2\left[\vec{a}\,\vec{b}\,\vec{c}\right]\vec{d}$

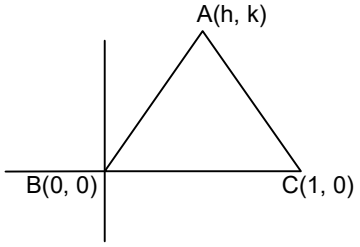
$$\left(\left(\vec{b}+\vec{c}\right)\times\left(\vec{b}+\vec{d}\right)\right)\times\left(\vec{c}\times\vec{d}\right)\cdot\vec{c} = \left(\left(\vec{c}\times\vec{b}+\vec{c}\times\vec{d}+\vec{b}\times\vec{d}\right)\times\left(\vec{c}\times\vec{d}\right)\right)\cdot\vec{c}$$

$= \left(\left[\vec{c}\,\vec{b}\,\vec{d}\right]\vec{c} - \left[\vec{b}\,\vec{d}\,\vec{c}\right]\vec{d}\right)\cdot\vec{c} = \left[\vec{b}\,\vec{d}\,\vec{c}\right]|\vec{d}|^2$.

47. AB

48. AD

Sol. B = 2C



$$\tan B = \frac{2\tan C}{1-\tan^2 C};\ \frac{k}{h} = \frac{\frac{2.k}{1-h}}{1-\frac{k^2}{(1-h)^2}}$$

$$\frac{\left(h-\frac{2}{3}\right)^2}{\frac{1}{9}} - \frac{k^2}{\frac{1}{3}} = 1;\ e = 2; \qquad \text{L. R} = 2$$

SECTION – C

49. 00008.00

Sol. Let α be the acute angle between AD and BE.

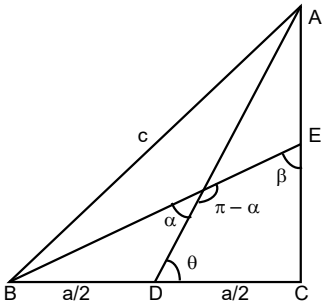
Then $\tan\alpha = \left|\frac{2-1}{1+2}\right| = \frac{1}{3}$.

Now $\theta + 180° - \alpha + \beta + 90° = 360°$

$\alpha = \theta + \beta - 90°$

$\tan\alpha = -\tan(\theta + \beta)$

$$\Rightarrow -3 = \frac{\tan\theta+\tan\beta}{1-\tan\theta\tan\beta} = \frac{\frac{2b}{a}+\frac{2a}{b}}{1-\frac{2b}{a}\cdot\frac{2a}{b}} = \frac{2b^2+2a^2}{-3ab} \Rightarrow 9ab = 2c^2.$$



50. 00024.00

Sol. $$f'(x) = 2.3^x.\ln 3\left[3.3^{2x} - 4.3x + 2\right]$$

$$= 3.2.3^x \ln 3\left[\left(e^x - \frac{2}{3}\right)^2 + \frac{2}{3}\right] > 0$$

In $[-1, 1]$

Hence

$f(x)$ is greatest when $x = 1 \& f(1) = 24$

51. 00002.72

Sol. $(x-1)(x-n) < 0$

$x \in (1, x)$

$\therefore x - 2x - 11 \qquad \Rightarrow x = 9$

$\cot^{-1}(\cot 9) = 9 - 2\pi$

52. 00003.14

Sol. $f'(x) = 3x^2 + 3 > 0$

$\Rightarrow \uparrow$ function and has exactly one real root $f(0) = \tan 2 < 0$

$\therefore \alpha \in (0, 1)$

$f(1) = 4 + \tan 2 > 0$

$\therefore \tan^{-1}\alpha + \tan^{-1}\dfrac{1}{\alpha} + \dfrac{\pi}{2} = \dfrac{\pi}{2} + \dfrac{\pi}{2} = \pi$

53. 00000.86

Sol. Orthocentre is foot of perpendicular drawn from origin on the plane.

54. 00004.00

Sol. $T_r = \dfrac{r^2}{(2r-1)(2r+1)} = \dfrac{1}{4}\left\{\dfrac{r}{2r-1} + \dfrac{r}{2r+1}\right\}$

$\therefore S = \dfrac{1}{4}\sum_{r=1}^{500}\left\{\dfrac{r}{2r-1} + \dfrac{r}{2r+1}\right\}$

$\therefore [S] = \dfrac{501000}{1001}$

$\therefore [S] = 125$

Total number of divisors = 4.